

Theories of Aging

Section 1 Introduction

The actual life span of any single individual is determined by a multitude of factors including lifestyle, socioeconomic status, diet, environmental conditions, and genetic endowments. The most dramatic effect of diet on lifespan is illustrated by the lifespan of the queen bee, compared to the worker bee. The queen bee can live as long as 6 years whereas worker bees live from 3 to 6 months, yet both the queen bee and the worker bee have the identical genetic makeup. The difference in lifespan is simply the result of whether or not the insects were fed royal jelly during their larva stage, which increases their lifespan by some twenty-fold.

As an organism ages, its chances of death increases, so that all individuals of a given species are dead by some age, characteristic of that species. It is estimated that the maximum lifespan for humans is between 90 – 120 years, although authentic reports of an individual living to be 122 years has been recorded. Theories to explain the mechanism of aging are divided into two main categories:

1. Developmentally-programmed aging
2. Damage accumulation aging

Programmed theories view aging as the result of genetic regulation (e.g. telomere shortening, hormonal changes etc). For example, with each cell division the telomeres (protein-DNA structures at the ends of the chromosomes in each cell) undergo shortening. At a critical level of telomere shortening cells show accelerated aging and an inability to keep dividing and a greater tendency to undergo cancerous changes as well as a greater susceptibility to free radical damage.

Damage theories view aging as the result of accumulated errors, primarily from free radicals. The two theories are complementary in that changes in gene expression that occur with cellular aging permit damage to occur to somatic cells that do not occur (or accumulate) in immortal cells such as germ cells (immature spermatozoa in the testes and the immature egg cells in the ovaries). In other words, aging does not occur in the germ cells, yet it clearly does so beginning sometime after fertilization. Thus, free radical aging is only allowed to begin (as determined by some genetic programming) only after fertilization has occurred. Evidence indicates that not only does free radical production increase substantially as we age, but the free radicals produced are less well contained and less efficiently mopped up by antioxidants, and their consequent damage is less efficiently repaired

Section 2 Developmentally-Programmed Aging

A. Telomeres

The ends of chromosomes consist of specialized nucleoprotein structures called telomeres. Telomeres are protein DNA structures present at the ends of chromosomes and are essential for genetic stability and cell replication. With each cell division, telomeres shorten until they reach a certain critical length, which causes cells to undergo senescence (cell cycle arrest).

Human telomeres function as a protective structure capping both ends of the chromosome. Uncapped chromosome ends are at risk of degradation, recombination, or fusion by cellular DNA repair systems, resulting in genetic information becoming lost, rearranged, or unstable. Telomeres have been shown to protect the ends of chromosomes from damage (free radicals) allowing for more complete replication of the chromosome, controlling gene expression and aiding in the organization of the chromosome. At cell replication, telomeres cannot be completely replicated. They are gradually shortened, and when the telomeres reach a critical threshold, cell replication is arrested in what is called "replicative senescence. Telomere length serves as an intrinsic biological clock regulating the life span of the cell, in which a finite number of cell replications can occur, upon which the cell undergoes replicative senescence.

To illustrate this point the human fibroblasts from young persons have long telomeres, whereas older individuals have shorter telomeres. Similarly, patients with premature aging syndrome (eg, Werner's syndrome and Hutchinson-Gilford's syndrome) have shorter telomeres as compared with healthy individuals of the same age. Thus, telomeres limit the capacity of a cell to replicate, by inducing senescence as a sort of tumor-suppressing mechanism. As discussed, with successive cell divisions, telomeres are gradually shortened. When they reach a critical length, the induction of senescence is associated with the activation of a number of pathways. As a consequence, cell growth is arrested and apoptosis (programmed cell death) occurs. However, during malignant transformation many cancer cells develop immortality by re-establishing telomere length via the secretion of telomerase enzyme.

Of clinical and diagnostic significance, with respect determining biological age of an individual, is the discovery that genetic determination of the regulation of telomere length is tissue-independent. Results indicate that telomere length measurement in easily accessible tissues such as blood cells could serve as a surrogate parameter for the relative telomere length in other tissues.

Long-lived species (humans) have cells with a far greater division potential than those of short-lived species like mice (humans live 5X longer than cats, who live 5X longer than mice, who live 25X longer than fruit flies)

Note that older people's cells in culture putter out much faster than those of younger people. Cells of a fetus (in culture) divide quickly until they too, eventually slow down and putter out.

At conception, telomere length is about 10,000 base pairs (compared to 130,000 base pairs for entire chromosome). At birth telomere length is about 5,000 base pairs

Telomerase Enzyme

Telomere length is usually maintained by a ribonucleoprotein called telomerase, which adds new DNA ends to chromosomes. This enzyme consists of a strand of RNA, called hTR (or hTerc), and a catalytic protein component, called hTERT. During human development, telomerase activity is detected in almost all tissues, while it is repressed in most normal adult human somatic tissues. In adults, telomerase is still expressed in the germline and in stem cells of renewable tissues such as bone marrow, skin, and the gastrointestinal tract.

Telomerase activity is almost absent in somatic cells, but it is detected in embryonic stem cells and in the vast majority of tumor cells. Tumor cells, in fact, may contain short and stable telomeres that confer immortality to the cancer cells, which are thus able to replicate indefinitely. The deregulation of telomeres thus plays an important role in the relationship between premature aging syndrome and cancer

An Historical Perspective Of Telomere and Telomerase Research

In the early 1930s, Hermann J. Muller and Barbara McClintock described the telomere (from the Greek word "telos," meaning end, and "meros," meaning part) as a protective structure at the terminal end of the chromosome. When this structure is absent, end-to-end fusion of the chromosome may occur, with ensuing cell death.

In the 1960s, Leonard Hayflick described a biological view of aging. He found that human diploid cells proliferate a limited number of times in a cell culture. The "Hayflick limit" is the maximal number of divisions that a cell can achieve in vitro. When cells reach this limit, they undergo morphologic and biochemical changes that eventually lead to arrest of cell proliferation, a process called "cell senescence". Hayflick noted that human fibroblasts could replicate a total of 50 times in culture, before no further cell division could occur. Near the final cell divisions, cells age more rapidly, look older, are larger and accumulate lipofuscin pigment and other debris.

In the 1970s, James D. Watson described what he called "end-replication problems." During DNA replication, DNA-dependent DNA polymerase does not completely

replicate the extreme 5' terminal end of the chromosome, leaving a small region of telomere uncopied, thus shortening the telomere with each cell division.

In the 1970s, Olovnikov connected cell senescence with end-replication problems in his "Theory of Marginotomy," in which telomere shortening was proposed as an intrinsic clocklike mechanism of aging that tracks the number of cell divisions before the arrest of cell growth or replicative senescence sets in. Greider and colleagues, in 1988, corroborated this theory when they observed a progressive loss in telomere length in dividing cells cultured in vitro.

In 1978, Elizabeth Blackburn found that the molecular structure of telomeres in *Tetrahymena pyriformis* contains long repeating units rich in thymine (T) and guanine (G) residues. In 1984, she and her colleagues isolated telomerase, the enzyme responsible for the maintenance and elongation of telomere length. In 1989, Gregg reported the existence of telomerase activity in human cancer cell lines, which was thought to contribute to the immortality of tumor cells. At about the same time, Greider and associates found that telomerase was nearly always absent in normal somatic cells.

In the 1990s, Shay and Harley detected telomerase in 90 of 101 human tumor cell samples (from 12 different tumor types), but found no activity in 50 normal somatic cell samples (from 4 different tissue types). Since then, more than 2600 human tumor samples have been examined and telomerase activity detected in about 90% of all tumor cells. The obvious implication is that telomerase may play a major role in the pathogenesis of cancer.

Because of their role in physiologic aging, cancer pathogenesis, and premature aging syndromes (eg, progeria), telomeres and telomerase are currently under intensive investigation.

Cancer and Telomerase Expression

Stem cells express telomerase to maintain telomere length throughout their life cycle. Many cancer cells have also been shown to express active telomerase. This might be a mechanism for their ability to avoid senescence and become immortal. About 90% of cancer cells contain short telomeres and high levels of telomerase activity. Alan Meeker, of the Johns Hopkins University School of Medicine in Baltimore, Maryland, has been investigating whether telomere shortening is actually a causative factor in tumorigenesis, or if it is simply a result of continuous cancer cell proliferation. Extensive research has led Meeker to conclude that telomere shortening is a prevalent feature of human premalignant epithelial cells. This supports the hypothesis that telomere shortening is an early event in human tumor development, and that such shortening could contribute to the chromosome mutagenesis and instability that are required for tumor progression. For example, 75% of oral carcinomas, 80% of lung cancers, 84% of prostate cancers, 85% of liver cancers, 93% of breast cancers, 94% of neuroblastomas, 95% of colorectal cancers,

and 98% of bladder cancers have detectable telomerase activity. It remains to be determined whether telomerase may represent a sensitive biomarker for diagnosis, early detection, and prognosis of cancer. Since telomerase activity appears necessary for the sustained proliferation of most cancer cells, anticancer agents based on telomerase inhibition might provide an effective therapy.

Nutrition Can Speed Up Or Slow Down The Rate Of Cell Replication

Cellular and tissue aging, as well as increased risk of cancer are clearly related to the shortening of telomeres with each cell division. Thus, any strategy that slows the rate of cell division prolongs the time in which an organism (including humans) can enjoy the benefits of healthier, more functional cells in all tissues and organs of the body, lower risk of developing cancer, and in effect, slow the aging process. Certain nutritional factors have been shown to either speed up or slow down the rate of cell division. Factors that speed up the rate of cell division, and thus, are associated with more rapid aging, and increased risk of cancer include:

1. A high animal fat diet - High levels of arachidonic acid give rise to the formation of prostaglandin series 2 hormones, which increase the rate of cellular replication throughout the body. The consumption of animal fat also increases the secretion of bile acids into the intestinal tract from the gallbladder and liver. Bile acids are readily converted by gut bacteria into secondary sterols, such as lithocholic acid and deoxycholic acid. These secondary sterols are known to increase the cell division rate of colonic epithelial cells and are associated with increased risk of colon cancer via this action.

2. Alcohol - Alcohol speeds up the rate of cell division, generates free radicals and acts as a co-carcinogen by enhancing the uptake of other carcinogens at the cellular level.

3. Obesity - Overweight subjects convert more androstenedione into estrone hormone within their fat tissue as a consequence of having larger fat cells. Estrone hormone increases cellular replication of breast and other female reproductive tissues and is strongly associated with increased breast cancer risk in postmenopausal women. Overweight postmenopausal women have a three times greater-risk of breast cancer than their leaner age-matched counterparts. Hormone replacement therapy increases risk of breast cancer in much the same way. In men, higher estrone levels inhibit the breakdown of dihydrotestosterone in the prostate gland. Dihydrotestosterone increases prostate cell division rates and is strongly associated with prostate enlargement and prostate cancer. Due largely to the consequence of creating insulin resistance, being overweight also inhibits the secretion of sex hormone binding globulin from the liver, which in turn, allows more estrogen and testosterone to circulate unbound to this important protein carrier, which allows these hormones to exert stronger physiological effects on reproductive tissue, speeding up of cell division rates with an associated increased risk of reproductive cancers.

Factors that slow down the rate of cell division and are thus, associated with a slowing of biological aging and reduced risk of cancer include:

1. Omega-3 fats – Omega-3 fats from fish, fish oil and flaxseed oil give rise to prostaglandin series-3, which slows cell division rates throughout the body.
2. Vitamin D – Stimulation of vitamin D receptors slows the rate of cell division. Higher circulating levels of vitamin D are associated with a decreased risk of colon, breast and prostate cancers. Calcium appears to work synergistically with vitamin D in regards to slowing the replication rate of colonic epithelium and thereby, reducing risk of colon cancer.
3. Vitamin A – Vitamin slows the rate of cell division and allows cells to fully mature (differentiate) in a highly regulated fashion, both effects of which are associated with reduced cancer risk
4. Nutrients That Inhibit The Synthesis Of Estrone Hormone In Fat Cells – As mentioned above, over secretion of estrone hormone from fat cells directly or indirectly speeds replication of breast, prostate and other reproductive tissues. Certain nutrients can block the aromatase enzyme responsible for the conversion of androstenedione into estrone hormone and thus, lower the circulating levels of this hormone. These protective agents include soy isoflavones (genistein and diadzein), the indole-3- carbinol from cruciferous vegetables (broccoli, cabbage, Brussels sprouts, cauliflower, bok choy) and the phytoestrogens (enterolactone and enterodiol) produced by the gut bacteria upon ingestion of flaxseed powder.
5. Bioactive Agents That Inhibit The Conversion Of Testosterone To Dihydrotestosterone In The Male Prostate Gland – A build up of dihydrotestosterone in the male prostate gland causes more rapid cellular replication leading to prostate enlargement and possibly increased risk of prostate cancer. Soy isoflavones (especially genistein), zinc, the bioactive agents in saw palmetto, and beta-sitosterol supplementation, have all been shown to inhibit the 5-alpha Reductase enzyme that converts testosterone into dihydrotestosterone, helping to slow prostate cell division rates.

Summary

Somatic cells are programmed to undergo only a limited number of cell divisions, after which they enter a state called senescence. This is most readily apparent in vitro, where cultured primary cells reach the end of their programmed lifespan, enlarge, and permanently cease to divide.

During our lifetime, as liver cells, kidney cells and the cells of other organs use up their maximum number of cell divisions and enter a state of senescence, fewer and fewer functional cells remain within any given organ as we age. As such, there is an age-related decline in functional capacity that affects most of the organ systems in the body due to

the fact that each organ will possess a fewer number of functional cells as time goes by, as individual cells reach their maximum replication potential. Thus, the ability of the kidneys to filter toxins from the blood becomes less efficient over time, as does the ability of the liver to detoxify blood toxins and the ability of the pancreas to secrete optimal levels of digestive enzymes etc. In short, much of the functional decline in the performance of the body's organ systems (which can lead to organ failure and death in the case of liver and/or kidney failure), is related to a loss of functional cells as the body ages. Therefore, any dietary or supplementation strategy that slows the rate of cell division of somatic cells, in theory, slows aging of internal organs and the body as a whole, and provides a longer period within one's lifespan in which these organs can remain more highly functional.

Primary cells that are driven to proliferate by oncogenic transformation bypass senescence and enter crisis, or massive cell death, as they approach the end of their lifespan. The small number of cells that survive crisis are capable of an infinite number of cell divisions, essentially becoming immortal cell lines. These in vitro studies have provided a model to study the biological clock of normal somatic cells, one of the most potent safeguards against malignant transformation and one that must be overcome by all cancer cells destined to grow past their prescribed lifespan.

The cellular biological clock is linked to telomeres, the ends of chromosomes that are necessary to maintain chromosomal integrity. Telomeres are composed of a stretch of repeated nucleotides. Because DNA polymerase fails to copy the extreme ends of chromosomes telomeres shrink with every cell division. Cellular senescence is well correlated with the shortening of telomeres to less than the size required to maintain the integrity of chromosomes. In contrast to somatic cells that have a finite lifespan, germ cells and hematopoietic stem cells preserve the length of their telomeres with an enzyme called telomerase. A ribonucleoprotein with reverse transcriptase activity, telomerase uses an RNA template to add the appropriate nucleotide repeats to the ends of telomeres, thus matching the progressive shortening that accompanies cell division. This physiologic mechanism for circumventing senescence has been adapted by human tumors, most of which express high levels of telomerase.

Stress Also Decreases Telomere Length and Speeds Cellular Aging

Chronic psychological stress is associated with accelerated telomere shortening and lower telomerase activity in leukocytes, which may explain how stress may ultimately lead to premature aging, according to a report in the Proceedings of the National Academy of Sciences. A recent study on the relationship between stress and aging, included a cohort of 39 healthy, premenopausal women who were primary care givers for a child with a chronic illness. Control subjects were 19 age-matched mothers of healthy children. Stress was measured with a standardized 10-item questionnaire in which the women reported their perceived level of stress over the past month. Objective stress was determined using the number of years since a child's diagnosis. Mean telomere length and telomerase

activity were measured in peripheral blood mononuclear cells. Oxidative stress was assayed in samples of serum and urine. Within the care giving group, an increased number of years as a caregiver was associated with shorter telomere length, lower activity of telomerase, and greater oxidative stress. The same relationship was observed between subjective stress and the markers of cellular aging. In the 14 women who comprised the highest quartiles of perceived stress scores, telomeres averaged 3,110 base-pairs in length, while the 14 in the lowest-stress quartile averaged 3,660 base pairs.

Of interest is the fact that during adulthood, telomere shortening averages 31 to 63 base pairs per year, so the authors estimate that the 550-base pair shortening in the high-stress group translates to 9 to 17 additional years of aging. These findings may have implications for human health, since telomere shortening and reduced telomerase are associated with premature death from cancer, cardiovascular and infectious disease. While the number of years that mothers had been a care giver did matter, "not all caregivers fell into the high-stress group," researchers noted. This implies that the ability to better manage stress throughout one's lifetime appears to result in a slower reduction in telomere length, and increased telomerase activity, which translates into slower biological aging and reduced risk of certain degenerative diseases.

References For Telomeres

1. Greider CW, Blackburn EH. Telomeres, telomerase and cancer. *Sci Am.* 1996;2:92-97.
2. Shay JW, Wright WE. Hayflick, his limit, and cellular ageing. *Mol Cell Biol.* 2000;1:72-76.
3. Reddel RR. The role of senescence and immortalization in carcinogenesis. *Carcinogenesis.* 2000;21:477-484.
4. Karp G. DNA replication and repair. In: *Cell and Molecular Biology*. 2nd ed. New York, NY: John Wiley & Sons, Inc. 1999;575-607.
5. Hao Y-h, Tan Z. The generation of long telomere overhangs in human cells: a model and its implication. *Bioinformatics.* 2002;18:666-671.
6. Makarov VL, Hirose Y, Langmore JP. Long G tails at both ends of human chromosomes suggest a C strand degradation mechanism for telomere shortening. *Cell.* 1997;88:657-666.
7. Griffith JD, Comeau L, Rosenfield S, et al. Mammalian telomeres end in a large duplex loop. *Cell.* 1999;97:503-514.
8. Blasco MA. Mammalian telomeres and telomerase: why they matter for cancer and aging. *Eur J Cell Biol.* 2003;82:441-446. [Abstract](#)
9. Greider CW. Telomeres do D-loop-T-loop. *Cell.* 1999;97:419-422.
10. Shay JW. At the end of the millennium, a view of the end. *Nat Gen.* 1999;23:382-383.
11. Kim S-H, Patrick K, Judith C. TIN2, a new regulator of telomere length in human cells. *Nat Gen.* 1999;23:405-412.
12. Shay JW, Zhou Y, Hiyama E, et al. Telomerase and cancer. *Hum Mol Gen.* 2001;10:677-685.

13. Enrique S, Fermin AG, Predrag S, et al. Mammalian Ku86 protein prevents telomeric fusions independently of the length of TTAGGG repeats and the G-strand overhang. *EMBO Rep.* 2000;11:244-252.
14. Bachand F, Ibtissam T, Autexier C. Human telomerase RNA-protein interactions. *Nucleic Acids Res.* 2001;29:3385-3393.
15. Greider CW. Telomerase activity, cell proliferation, and cancer. *Proc Natl Acad Sci U S A.* 1998;95:90-92.
16. Haber DA. Clinical implications of basic research: telomeres, cancer and immortality. *N Engl J Med.* 1995;332:955-956.
17. Belair CD, Yeager TR, Lopez PM, et al. Telomerase activity: a biomarker of cell proliferation, not malignant transformation. *Proc Natl Acad Sci U S A.* 1997;94:13677-13682.
18. Bodnar AG, Ouellette M, Frolkis M, et al. Extension of life-span by introduction of telomerase into normal human cells. *Science.* 1998;279:349-352.
19. Cong Y-S, Wright WE, Shay JW. Human telomerase and its regulation. *Microbiol Mol Biol Rev.* 2002;66:407-425.
20. Johnson FB, Sinclair DA, Guarente L. Molecular biology of aging. *Cell.* 1999;96:291-302.
21. Wright WE, Shay JW. Telomere dynamics in cancer progression and prevention: fundamental differences in human and mouse telomere biology. *Nat Med.* 2000;6:849-851.
22. Stewart SA, Hahn WC, O'Connor BF, et al. Telomerase contributes to tumorigenesis by a telomere length-independent mechanism. *PNAS.* 2002;99:12606-12611.
23. Judy MYW, Collins K. Telomere maintenance and disease. *Lancet.* 2003;362:983-988.
24. Simon WLC, Blackburn EH. New ways not to make ends meet: telomerase, DNA damage proteins and heterochromatin. *Oncogene.* 2002;21:553-563.
25. Herbert BS, Pitts AE, Baker SI, et al. Inhibition of human telomerase in immortal human cells leads to progressive telomere shortening and cell death. *PNAS.* 1999;96:14276-14281.
26. Halvorsen TL, Leibowitz G, Levine F. Telomerase activity is sufficient to allow transformed cells to escape from crisis. *Mol Cell Biol.* 1999;19:1864-1870.
27. Grogelny JV, McEliece MK, Broccoli D. Effects of reconstitution of telomerase activity on telomere maintenance by the alternative lengthening of telomeres (ALT) pathway. *Hum Mol Gen.* 2001;10:1953-1961.
28. Bryan TM, Englezou A, Dalla PL, et al. Evidence for an alternative mechanism for maintaining telomere length in human tumors and tumor-derived cell lines. *Nat Med.* 1997;3:1271-1274.
29. Liu JP. Studies of the molecular mechanisms in the regulation of telomerase activity. *FASEB J.* 1999;13:2091-2104.
30. Zhang XL, Mar V, Zhou W, et al. Telomere shortening and apoptosis in telomerase-inhibited human tumor cells. *Gen Dev.* 1999;13:2388-2399.
31. Hahn WC, Weinberg RA. Mechanisms of disease: rules for making human tumor cells. *N Engl J Med.* 2002;347:1593-1603.
32. Blasco MA, Lee HW, Hande MP, et al. Telomere shortening and tumor formation by mouse cells lacking telomerase RNA. *Cell.* 1997;91:25-34.
33. Blasco MA. Telomeres in cancer therapy. *J Biomed Biotechnol.* 2001;1:3-4.
34. Hodes R. Molecular targeting of cancer: telomeres as targets. *Proc Natl Acad Sci U S A.* 2001;98:7649-7651.
35. Mergny JL, Riou JF, Mailliet P, et al. Natural and pharmacological regulation of telomerase. *Nucleic Acids Res.* 2002;30:839-865.
36. Herbert BS, Wright WE, Shay JW. Telomerase and breast cancer. *Breast Cancer Res.* 2001;3:146-149.
37. Blasco MA. Mouse models to study the role of telomeres in cancer, aging and DNA repair. *Eur J Cancer.* 2002;38:2222-2228.
38. Cong YS, Wright WE, Shay JW. Human telomerase and its regulation. *Microbiol Mol Biol Rev.* 2002;66:407-425.
39. Meeker AK. Direct evidence of telomere shortening in the majority of human epithelial pre-cancerous lesions. Program and abstracts of the AACR Meeting on the Role of Telomeres and Telomerases in Cancer; December 7-10, 2002; San Francisco, California.
40. Meeker AK, Gage WR, Hicks JL, et al. Telomere length assessment in human archival tissues: combined telomere fluorescence in situ hybridization and immunostaining. *Am J Pathol.* 2002;160:1259-1268.
41. Reddel RR. Alternative lengthening of telomeres in human cell lines and tumors. Program and abstracts of the AACR Meeting on the Role of Telomeres and Telomerases in Cancer; December 7-10, 2002; San Francisco, California.
42. Neumann AA, Reddel RR. Telomere maintenance and cancer -- look, no telomerase. *Nat Rev Cancer.* 2002;2:879-884.
43. DePinho RA. Crisis in the telomere checkpoint response in normal and neoplastic tissues. Program and abstracts of the AACR Meeting on the Role of Telomeres and Telomerases in Cancer; December 7-10, 2002; San Francisco, California.
44. Maser RS, DePinho RA. Connecting chromosomes, crisis, and cancer. *Science.* 2002;297:565-569.
45. Chang S, Khoo CM, Naylor ML, Maser RS, DePinho RA. Telomere-based crisis: functional differences between telomerase activation and ALT in tumor progression. *Genes Dev.* 2003;17:88-100.
46. Weinberg RA. Control of replicative senescence by the telomere-associated single-strand overhang. Program and abstracts of the AACR Meeting on the Role of Telomeres and Telomerases in Cancer; December 7-10, 2002; San Francisco, California.
47. Stewart SA, Hahn WC, O'Connor BF, et al. Telomerase contributes to tumorigenesis by a telomere length-independent mechanism. *Proc Natl Acad Sci U S A.* 2002;99:12606-12611.
48. Harley CB. Status and issues for therapeutics based on telomere biology. Program and abstracts of the AACR Meeting on the Role of Telomeres and Telomerases in Cancer; December 7-10, 2002; San Francisco, California.
49. Chin AC. Treatment with GRN163 (novel telomerase template antagonist) selectively inhibits growth of myeloma and lymphoma cell lines and tumor xenografts by inducing proliferative arrest and apoptosis in malignant cells with short telomeres. Program and abstracts of the AACR Meeting on the Role of Telomeres and Telomerases in Cancer; December 7-10, 2002; San Francisco, California.
50. Irving J. An oncolytic adenovirus controlled by the human telomerase promoter provides broad-spectrum anti-tumor activity. Program and abstracts of the AACR Meeting on the Role of Telomeres and Telomerases in Cancer; December 7-10, 2002; San Francisco, California.
51. Novak KD. Telomeres and telomerase in cancer: Highlights From an AACR Special Conference; December 7-11, 2002; San Francisco, California. *Medscape General Medicine* 5(1), 2003. © 2003 Medscape Posted 01/23/200

52. Telomere Length in Different Tissues of Elderly Patients. Friedrich U, Griesse E, Schwab M, Fritz P, Thon K, Klotz U Mech Ageing Dev. 2000;15:89-99
53. Proc Natl Acad Sci USA 2004. www.pnas.org/cgi/doi/10.1073/pnas.040716210

B. The NeuroEndocrine Clock

Programmed into the body's aging clock are what appears to be predetermined points in the human lifecycle whereby hormonal changes are triggered, which permit the aging process to advance more rapidly. At specific points in the lifecycle, which are highly predictable, a distinct pattern of decline in hormone secretion occurs for melatonin, growth hormone, insulin-like growth factor-1, DHEA, estrogen, progesterone and thymosin (and other thymus gland hormones), with each hormone exhibiting its own unique pattern of decline. The age-related altered patterns of hormone synthesis and secretion are known to contribute to cellular aging of many target tissues and to many clinical signs and symptoms associated with the aging process of the body as a whole.

Melatonin- melatonin is synthesized in the pineal gland from serotonin. Babies produce increasing amounts of melatonin until they are about one year old. Then their nocturnal melatonin levels (melatonin is primarily secreted at night to induce sleep) remain steady until just before puberty, when they decline markedly, a trend that continues for the next five years or so. The abrupt decline just before puberty appears to be part of the neuroendocrine signal that triggers the onset of puberty. The onset of puberty is associated with increased release of reproductive hormones and growth hormone. Animal studies show that the administration of high levels of melatonin can inhibit reproduction. In children, those who are late to enter puberty have been found to have higher levels of melatonin than their peers, whereas individuals who enter puberty as early as three years before their peers may produce only one-third as much melatonin as other children their age.

Melatonin continues to decline from just before the onset of puberty through to age 70, such that a person at age 9 or 10 years of age exhibits nighttime melatonin blood levels at approximately 125 pg/ml compared to a 70 year old whose nighttime blood levels are customarily 10-20 pg/ml. Melatonin is the hormone that induces deep levels of sleep, helping the body to renew itself over night. This hormone is also important to regulate and optimize immune function. The decline in melatonin secretion corresponds with the time in our lifespan that is associated with the initial stages of shrinking of the thymus gland, which is an important immune system gland. T-helper cells of the immune system (a type of white blood cell) have been shown to have melatonin receptors. Upon

stimulation by melatonin these immune cells release a number of signaling chemicals that heighten the performance of many other immune cells and immune system activities. Melatonin is also a powerful antioxidant, which protects the brain from free radical damage (and other tissue sites as well). Melatonin has shown impressive anti-cancer and tumor suppressive effects in animal studies and preliminary human trials. Animal studies have shown that it can increase longevity of mice by up to 20% when provided in their nightly drinking water, compared to mice not receiving melatonin during their lifespan. The decline in melatonin secretion is considered to be an important hormonal factor in the aging process.

Growth Hormone (GH) – Growth hormone (somatotropin) is a single-chain peptide of 191 amino acids, and is the most abundant hormone in the body. It is produced by the somatotroph cells in the anterior pituitary gland. Growth hormone (GH) declines in every animal species that has been tested to date. In humans the amount of GH secreted after age twenty-one to thirty falls about 14% per decade, such that the total twenty-four-hour GH production rate is reduced by 50%, by the age of sixty. This means that at twenty years old we make about 500 mcg per day, 200 mcg at forty years old and only 25 mcg at 80 years old. Once secreted by the anterior pituitary gland, growth hormone is extracted from the blood stream by the liver where it stimulates the synthesis and release of insulin-like growth factor-1, which produces the actual effects on our tissues that are attributed to growth hormone. Insulin-like growth factor-1 (IGF-1) has an anabolic effect on lean mass and bone mass, and thus, speeds metabolism indirectly by encouraging lean mass development. IGF-1 also strengthens immune function and appears to positively impact libido. The age-related decline in GH and IGF-1 greatly contribute to the age-related loss in lean mass (muscle catabolism) and bone loss, a gain in fat mass, and has been shown to contribute to many other aspects of aging, which may include decreased energy, thinning hair, increased wrinkles, sagging and loose skin, and frail nails

DHEA – DHEA (dehydroepiandrosterone) is produced by the adrenal glands upon stimulation of ACTH (released by the pituitary gland), using cholesterol as a precursor. DHEA is also produced in the ovaries and testis. DHEA is called the “the mother of all steroids”, and is the most abundant steroid hormone in the body. It is the parent compound from which the body makes testosterone, estrogen, progesterone and corticosterone. The body’s internal clock alters the secretion rate of DHEA as we age, which in turn, has far reaching effects on the aging process. The decline in DHEA secretion parallels that of growth hormone. Blood levels of DHEA peak at about age 20 (men are usually in the range of 320-380 mcg/dl with women in the range of 200-250 mcg/dl), and begin to decline at a steady gradual rate after thirty. By age 75, many people produce 75% less DHEA than they did at age 20. The indirect effects of DHEA decline are usually first experienced between age 40 and 50. Because DHEA is the raw material from which the body makes other steroid hormones, the drop off in DHEA production impacts the synthesis of testosterone, estrogen and progesterone, which in turn, have profound effects on the aging process of the body.

Testosterone – The decline in DHEA production results in a decreased synthesis of testosterone production by the testis and adrenal glands as men age. Studies indicate that

there is a 1% decline in testosterone production each year between age 40 and 70, such that a 70-year old man often shows a 30% reduction in blood testosterone levels compared to a 40-year old man. Testosterone levels continue to drop off after age 70 as well. The decline in testosterone production is strongly linked to many age-related symptoms and complaints including reduced libido and sexual performance capabilities (erectile or sexual dysfunction), fatigue, depression, irritability, loss of lean mass (catabolism), increased fat gain, and loss of bone mass after age 65. In women, after menopause testosterone levels are reported to decline as much as 50%. The decline in testosterone levels in menopausal and postmenopausal women has been reported to contribute to low sex drive as well as age-related bone loss

Estrogen – Beginning at menopause there is a marked reduction in estrogen synthesis by the ovaries in women, such that a 90% reduction in estrogen blood levels are a routine finding by the end of the menopausal period. This occurs due to the fact that ovulation is no longer occurring in a menopausal and postmenopausal woman, as it is the developing and ovulated egg cell that produces the majority of estrogen in the adult body of a woman. The decline in estrogen secretion after age 50 encourages loss of calcium from bone, an easier build of cholesterol in the blood stream (due to decreased LDL-cholesterol receptor synthesis) atrophy of reproductive tissue, reduced libido, and accelerated aging of the skin, hair and other tissues.

Conversely, in men, there is usually an increase in estrogen output by fat tissue as men age, which contributes to the age-related increase in prostate enlargement, and increased fat mass deposition in the abdominal region.

Progesterone – Like estrogen, there is a marked decline in progesterone production during menopause such that women often exhibit a 66% decline in blood progesterone levels by the end of the menopausal period. This age-related decline in progesterone has been shown to encourage bone loss, loss of libido, and atrophy of reproductive tissue in women.

Thymus Gland Involution and Immune Function – Peak function of the body's immune system is reported to occur around puberty and gradually declines with advancing age. The decline in immune function is largely attributed to the age-related shrinking of the thymus gland. After the teen years the thymus gland begins to shrink (undergo involution) until by age 40 it is a shriveled shadow of its former self. By age sixty the thymus gland is difficult to locate due to the significant degree of involution that has taken place. The thymus gland produces T lymphocytes, T cells essential for the resistance to infection and protection against the development of cancer and autoimmune diseases. The thymus gland also produces a number of hormones (e.g. thymosin), which are important modulators of immune function. The release of these hormones decline as the thymus gland shrinks with advancing age. The shrinkage of the thymus gland coincides with a rise in the diseases associated with aging, including cancer, autoimmune diseases, and infectious diseases. There is also a decline in the release of immune system signaling factors (interleukins and cytokines), which further contributes to the age-related decline to immune surveillance and immune function.

Summary

The age-related decline in key hormones mentioned above is known to play an important role in the aging process of the body. Attempts to stave off age-related changes associated with hormonal decline have employed the use of melatonin supplementation, hormone replacement therapy (estrogen plus progesterone), growth hormone injections and/or the use of oral growth hormone secretagogue supplements, and DHEA supplementation. Some of these interventions have been shown to increase incidence of certain cancers while others appear to be safe and highly effective at slowing and reversing some aspects of the aging process and defending against certain degenerative diseases.

References For The Neuroendocrine Clock

1. Reiter R. Melatonin. Bantam Books. 1995: 21-29
2. Klatz R and Goldman R. The New Anti-Aging Revolution. Basic Health Publications, Inc. 2003: 19-38
3. Klatz R. Grow Young With HGH. HarperPerennial. 1998: 39-48, 176-199
4. Rouzier N. Natural Hormone Replacement. WorldLink Medical Publishing. 2001: 17-40, 49-62, 63-88, 89-106, 107-152, 181-208
5. Shelian R. DHEA: A Practical Guide. Avery Publishing Group. 1996
6. Sahelian R. Melatonin: Nature's Sleeping Pill. Avery Publishing Group. 1995

Section 3 Damage Accumulation Theories

A. Free Radicals, Antioxidants and Aging

Free radicals produced during normal metabolism cause damage to macromolecules. The free radical theory of aging proposes that the organism is unable to repair all of the free radical damage created and that, with time, un-repaired damage accumulates and puts the organism at risk: in other words, free radicals provoke cellular aging and promote biological changes within the organism that are associated with degenerative states and increasing risk of death. In addition to exogenous free radicals such as cigarette smoke, alcohol consumption, other pollutants and food-borne free radicals, many free radicals

are reactive oxygen species that are byproducts of endogenous oxidation reactions. Free radical damage has been implicated in the pathogenesis of several diseases. According to the free radical theory of aging, free radicals are involved in the production of changes in cellular metabolism that lead to a time-dependent functional decline in all living beings. Consequently, antioxidant and/or free radicals scavengers may retard the aging process. Severe oxidative stress progressively leads to cell dysfunction and ultimately cell death. Oxidative stress is defined as an imbalance between pro-oxidants and/or free radicals on the one hand, and anti-oxidizing systems on the other. The oxygen required for living may indirectly be responsible for negative effects; these deleterious effects are due to the production of free radicals, which are toxic for the cells (superoxide anions, hydroxyl radicals, peroxy radicals, hydrogen peroxide, hydroperoxides and peroxynitrite anions). Free radical attacks are responsible for cell damage and the targeted cells are represented by the cell membranes, which are particularly rich in unsaturated fatty acids, sensitive to oxidation reactions; DNA is also the target of severe attacks by these reactive oxygen species (ROS). The cell's defense system is comprised of antioxidant enzymes and antioxidant nutrients, which include such components as vitamins (E and C), enzymes [superoxide dismutase (SOD), glutathione peroxidase (GPx) and others], and glutathione reductase (GSH), capable of neutralizing peroxynitrite. The efficacy of this system is dependent on the genome for the enzymatic defense systems, and on nutritional antioxidants. In addition to containing free radicals associated with aging, epidemiological data suggest that antioxidants may have a beneficial effect on many age-related diseases: atherosclerosis, cancer, some neurodegenerative and ocular diseases. For instance, the post-secretory oxidation of LDL by oxygen free radicals results in the unregulated uptake of cholesterol in arterial walls, which accelerates the atherosclerotic process. Cumulative oxidative damage to DNA may be associated with the rise in cancer risk that occurs with increasing age. Excess production of free radicals may also contribute to tissue damage in rheumatoid arthritis, inflammatory bowel diseases (including Crohn's disease and ulcerative colitis), cataracts, macular degeneration, and neurodegenerative diseases and are strongly linked to the aging process. To fully understand the relationship between oxygen, free radicals, antioxidants, and their effects on aging and the disease process, a review of some evolutionary concepts and physiological principles should prove helpful.

Humans Evolved As Oxygen Consumption Machines

Millions of years ago when the earth was formed, there was no free oxygen on the planet. Then after a few millennia, one-celled creatures appeared that did not need oxygen to survive. Instead, they used energy derived from the sun in a primitive version of photosynthesis to draw carbon and nitrogen atoms directly from the atmosphere, which became the raw materials they needed to sustain life. After a while more sophisticated cells evolved that were able to use water as part of their photosynthesis process. A waste product of this process was oxygen. Gradually these primeval life forms multiplied and oxygen accumulated in the atmosphere. The process of photosynthesis converted an atmosphere with virtually no free oxygen to an atmosphere that now is comprised of 21% oxygen content. The original anaerobe creatures were now at a disadvantage, as they had no protection against oxygen, which is highly toxic to an organism that has no

antioxidant defenses (antioxidant enzymes, melatonin, alpha-lipoic acid and uric acid). Natural selection now favored the appearance of organisms that could use oxygen in their energy production pathway and at the same time protect themselves from the corrosive effects of oxygen.

As such, more sophisticated oxygen-dependent organisms evolved (including humans) that use oxygen in the conversion of macronutrients (carbohydrates, fat, protein, alcohol) to ATP energy, which powers the biological machinery of the body. In human aerobic metabolism, carbohydrates, protein, fat and alcohol are first converted to acetyl Coenzyme A, which combines with oxaloacetate to enter the Krebs cycle, forming citrate. During the many reactions within the Krebs cycle hydrogen atoms are removed from the various intermediate compounds by niacin (NAD) and riboflavin (FAD). The hydrogen atoms are then escorted to the electron transfer chain within the mitochondria (the place where aerobic metabolism occurs) and the electrons from hydrogen are then transferred from one cytochrome to the next until they reach the last cytochrome (cytochrome oxidase). At that point 4 hydrogen electrons that have been shuttled down the cytochrome chain react with one oxygen di-radical (two oxygen atoms attached together written as O^2) to form H^2O (water). As the hydrogen electrons are passed from one cytochrome to the next they are moving from higher to lower energy states. The energy difference between the higher level of energy and the lower level of energy, as the electrons move down the electron transfer chain, is given off as free energy and used to re-couple ADP (adenosine monophosphate) with inorganic phosphate, to form ATP (adenosine triphosphate). When enzymes split the outer most phosphate off from ATP, it releases a large parcel of energy that is used to power the biological machinery of the body (muscle contraction, energy dependent enzyme reactions etc.). At this point ATP has lost a phosphate and is now in the form of ADP (adenosine diphosphate). As hydrogen electrons continue to be shuttled down the cytochrome electron transfer chain, ADP can once again be cycled back to ATP and the energy dependent reactions of the body continue. Thus, to keep the cells of the body alive, the conversion of carbohydrate, fat, protein (and when available, alcohol) to acetyl CoA, and the ultimate removal of hydrogen atoms from Krebs cycle intermediates to be shuttled down the cytochrome electron transfer chain to produce ATP, must continue at all times. If blood vessels become clogged with atherosclerotic plaque (usually secondary to hypercholesterolemia) then oxygen can no longer be delivered to the cells that are distal to the arterial blockage. With no oxygen in the cell to receive the hydrogen electrons being shuttled down the electron transfer chain, a backing up effect occurs in the electron transfer chain system, whereby hydrogen electrons can no longer be shuttled from cytochrome to cytochrome. The net result is that ATP can no longer be re-synthesized and cell death ensues due to an energy shortage within the cell. This is classically what happens to myocardial cells during a heart attack and in the brain during an ischemic stroke, cerebral vascular insufficiency or a transient ischemic attack.

Also note that as we age the synthesis of coenzyme Q10 declines. Coenzyme Q10 is required to shuttle hydrogen electrons from the first cytochrome to the second cytochrome within the cytochrome electron transfer chain. A decline in coenzyme Q10 concentrations results in fewer hydrogen electrons being transferred down the through the cytochrome system and the net result can be impaired ATP synthesis, which is linked to the development of congestive heart failure, high blood pressure, aggravation of chronic

stable angina and Parkinson's disease. This is why many anti-aging experts suggest taking Coenzyme Q10 supplementation by age 40-50.

So Where Do The Oxygen Free Radicals Come From?

As noted above, hydrogen electrons (originally derived from macronutrients) are continually shuttled from one cytochrome to the next within the electron transfer chain, as means to release energy that re-couples ADP with inorganic phosphate to synthesize ATP. This process is often referred to as oxidative phosphorylation. However, it is not a foolproof system in that 2-5% of the hydrogen electrons leak out of the system (within the cytochrome electron transfer chain). When they leak out of the system they become very reactive with the oxygen di-radical (O^2). Oxygen is missing two electrons in its outer most orbital shells and is seeking to fill these outer most electron shells with the missing electrons. When oxygen absorbs one hydrogen electron that has leaked out of the system, it is then converted into a free radical known as the superoxide anion, which now has one unpaired electron in its outer orbital shell. This is a highly reactive free radical that will randomly steal an electron from a near by molecule (DNA, cell membrane unsaturated fat, enzyme, etc.) in order to become electrically stable. However, the molecule that has had an electron stolen away now has an unpaired electron of its own and must, in turn, strip an electron away from another near by molecule, thereby converting it into a free radical. And so free radical attack will continue within the cell until the process is stopped by an antioxidant. Antioxidants can donate an electron to stabilize a free radical without becoming converted into a free radical itself. (or more correctly, via this process they quench more free radicals than they create, as some antioxidants can become pro-oxidants under certain conditions, but in general, antioxidants intercept free radicals and reduce free radical, oxidative stress to the cells of the body).

In certain instances, the superoxide anion may absorb an additional hydrogen electron that has leaked out of the system, which converts it into hydrogen peroxide (H^2O^2). Hydrogen peroxide is known to travel throughout the cell and diffuse through the nuclear membrane to reside in close proximity to the DNA of the cell. In the presence of a transition metal (iron, copper, some heavy metals) hydrogen peroxide is easily converted into hydroxy radicals (HO^* plus *OH) when the transition metal donates an electron to the reaction with hydrogen peroxide. In this instance the transitional metal becomes more positively charged (e.g. divalent iron; ferrous iron, becomes trivalent iron; ferric iron). The hydroxy radicals created in this instance are highly reactive oxygen free radicals that are known to produce DNA damage and to damage many other cellular components. The other oxygen free radical of importance is known as singlet oxygen, which is created when the oxygen within our cells absorbs additional energy from X-rays, and/or ultra-violet light (UV-light creates free radicals in the skin, and the lens and macula of the eye).

Thus, oxygen is a double-edged sword in that we need oxygen within our cells to allow the production of ATP to occur, and at the same time oxygen free radicals are continuously being formed as by-products of a leaky, inefficient oxidative phosphorylation system within the mitochondria of each cell, and the interaction of

cellular oxygen with ultra-violet light and X-radiation energy. According to Russel Reiter, a professor of neuroendocrinology at the University of Texas Health Science Center at San Antonio, each cell is exposed to one trillion oxygen free radicals alone, each day. Most of these free radicals are quenched by antioxidant enzymes, but not all of them. Slowly over time the cumulative damage created by free radicals leads to the kind of cellular damage associated with aging and degenerative diseases, including cancer. Consistent with this process, the mitochondria theory of aging contends that free radical damage to the mitochondria structure itself (the mitochondria is in close proximity to the area where most of the oxygen free radicals are first created) damages the mitochondria severely enough to reduce its capacity to generate ATP energy. The result is a power shortage within the cell, which in turn leads to further cellular dysfunction, aging and increased risk of cell death. Of interest to anti-aging practitioners is that coenzyme Q10, alpha-lipoic acid, and Vitamin E have been shown to protect the mitochondrial membrane and mitochondrial DNA, from free radical damage, by acting as antioxidants in this area of the cell. This is a further reason to consider coenzyme Q10 supplementation by age 40 and lifelong supplementation with Vitamin E. In terms of why some species age more slowly and live longer than others, some available data indicate that the relation between oxidative stress and aging is due to the presence of low rates of mitochondrial free radical production and low degrees of fatty acid unsaturation of cellular membranes in the post-mitotic tissues of long-lived animals in relation to those of short-lived ones. Recent research shows that long-lived animals also have lower steady-state levels of oxidative damage in the mitochondrial DNA (mtDNA) of post-mitotic cells than short-lived species. Studies also show that the flux of free radical attack to mtDNA is higher in short- than in long-lived animals, and suggest that this is a main determinant of the rate of accumulation of mtDNA mutations, and thus the rate of aging.

A Quick Review of Oxygen Free Radicals And Antioxidants In The Aging Process

As stated above there are four common oxygen metabolites in biologic systems that are free radicals: superoxide anion (O_2^-), hydrogen peroxide (H_2O_2), hydroxyl radical ($OH^\cdot$), and singlet oxygen (1O_2). These free radicals can be generated via a number of mechanisms, including normal physiologic processes and processes resulting from external factors. For example, singlet oxygen is generated by photosensitization reactions wherein a molecule absorbs light of a given wavelength, exciting the molecule. This excited molecule transfers the increased energy to molecular oxygen, creating singlet oxygen, which then can attack other cell components. It appears that the primary function of carotenoids, an important class of antioxidants, is to scavenge free radicals, particularly singlet oxygen produced by photosensitization.

A certain amount of oxidative function is necessary for proper health. For example, oxidation processes are used by the body's immune systems to kill microorganisms. Sometimes, however, the level of toxic reactive oxygen intermediates (ROI) overcomes the antioxidant defenses of the host, resulting in an excess of free radicals and a state called oxidative stress. These free radicals can induce local injury by reacting with lipids,

proteins, and nucleic acids. The interaction of free radicals with cellular lipids leads to membrane damage and the generation of lipid peroxide byproducts.

Cells contain a number of antioxidants that have various roles in protecting against free radical reactions. The major water-soluble antioxidant metabolites are glutathione (GSH) and vitamin C, which reside primarily in the cytoplasm and mitochondria. Many water-soluble enzymes also catalyze these reactions. Glutathione peroxidase catalyzes the reaction between GSH and hydrogen peroxide to form water and oxidized glutathione, which is stable.

Vitamin E and the carotenoids are the principal lipid-soluble antioxidants. Vitamin E is the major lipid-soluble antioxidant in cell membranes that can break the chain of lipid peroxidation. Therefore, theoretically, it is the most important antioxidant in preventing oxidation of these fatty acids. Vitamin E is recycled by a reaction with vitamin C.

Despite the actions of antioxidant nutrients, some oxidative damage will occur, and accumulation of this damage throughout life is believed to be a major contributing factor to aging and disease.

In addition to their importance to degenerative diseases and cellular aging, free radicals also appear to be responsible for the increase in cross-linking between collagen fibers within our body tissues as we age. One of the most common proteins found in the skin, tendons, ligaments, bone, and cartilage is collagen. Very youthful collagen exhibits widely spaced cross-linking between parallel collagen fibers (like rungs in ladder connecting the two main legs together). It appears that free radicals are responsible for the increased number of cross-links (increased number of rungs in the ladder) that occur in our collagen as we age. The increased cross-linking causes the skin to shrink, and our muscles, joints and tendons to become stiffer and less flexible and pliable, and reduces the expansive capacity of our lungs, which contributes to the pulmonary functional decline that accompanies aging. Thus, reduced free radical exposure and more superior status of antioxidants status may also reduce the rate of collagen cross-linking, which counters other important aspects of aging.

References For Free Radicals and Antioxidants

1. Meyers DG, Maloley PA, Weeks D. Safety of antioxidant vitamins. *Arch Intern Med.* 1996;156:925-35.
2. Sun Y. Free radicals, antioxidant enzymes, and carcinogenesis. *Free Radic Biol Med.* 1990;8:583-99.
3. Loudon GM. *Organic Chemistry*. Menlo Park, Calif: Benjamin/Cummings Publishing Company Inc; 1988.
4. Winkler BS, Boulton ME, Gottsch JD, Sternberg P. Oxidative damage and age-related macular degeneration. *Mole Vis.* 1999;5:32.
5. Bulger EM, Helton WS. Nutrient antioxidants in gastrointestinal diseases. *Gastroenterol Clin North Am.* 1998;27:403-19.
6. Machlin LJ, Bendich A: Free radical tissue damage: protective role of antioxidant nutrients. *FASEB J* 1987; 1:441-445

7. Ames BN, Shigenaga MK, Hagen TM: Oxidants, antioxidants, and the degenerative diseases of aging. *Proc Natl Acad Sci U S A* 1993; 90:7915-7922
8. Barja G The flux of free radical attack through mitochondrial DNA is related to aging rate. *Aging (Milano)* 2000 Oct;12(5):342-55
9. Bonnefoy M; Draï J; Kostka T Antioxidants to slow aging, facts and perspectives] *Presse Med* 2002 Jul 27;31(25):1174-84
10. Bonnefoy M; Draï J; Kostka T Oxidative stress, aging and longevity in *Drosophila melanogaster*. *FEBS Lett* 2001 Jun 8;498(2-3):183-6
11. Reiter R. Melatonin. Bantam Books. 1995: 15-44
12. Klatz R and Goldman R. The New Anti-Aging Revolution. Basic Health Publications, Inc. 2003: 19-38
13. Halliwell B and Gutteridge J. Free Radicals in Biology and Medicine (second edition) 1991: 1-20; 416-508

B Accumulated Glycosylated End-Products (AGE-Proteins)

In the presence of elevated blood glucose levels, many of the body's proteins become glycosylated (sugar coated by glucose or sorbitol). Virtually all proteins in the body that are rich in the amino acid lysine can undergo glycosylation as a consequence of the reaction of the epsilon amino group on lysine, with the aldose form of glucose, to form a glycosylated protein. Proteins such as serum albumin and hemoglobin rapidly undergo glycosylation when blood glucose levels are elevated. Glycosylated proteins are known to clog blood vessels, damage internal organs and contribute to cataract formation. The degree of protein glycosylation has been used as a marker for biological aging of specific organs and loss of reserve function. The best evaluation of recent glycosylated protein activity (previous 17-20 days) is measured by assessing fructosamine concentrations from blood samples. A normal reading is in the about 2.06 mmol/l, and when corrected for plasma proteins, 30.5 mmol/gram. Many anti-aging experts recommend keeping fasting blood levels at or below 90 mg/dl (5.0 mmol/l) as a proven means to reduce the accumulation of glycosylated proteins that contribute to organ damage (especially damaging to the kidney tubules), cardiovascular disease and cataract formation as one ages.

References For Glycosylated Proteins

1. Tahara Y, Shima K. Kinetics of HbA1c, glycated albumin, and fructosamine and analysis of their weight functions against preceding plasma glucose level. *Diabetes Care*. 1995;18(4):440-447.
2. Broussolle C, Tricot F, Garcia I et al. Evaluation of the fructosamine test in obesity: consequences for the assessment of the past glycemic control in diabetes. *Clin Biochem*. 1991;24:203-209
3. Kneht KJ, Dunn JA, McFarlane KF, et al. Effect of diabetes and aging on the carboxymethyllysine levels in human urine. *Diabetes*. 1991;40:190-196
4. Wu JT, Tu MC, Zhung P. Advance glycation end product (AGE): characterization of the products from the reaction between D-glucose and serum albumin. *J Clin Lab Analysis*. 1996;10(1):21-34

Theories of Aging And Nutrition

As described in this review, several theories of aging have been proposed over the years which highlight the involvement of both internal genetically timed events as well the accumulation of damage to cells that occurs from exposure to free radicals and glycosylated proteins (AGE-proteins). Studies suggest that certain dietary practices and the use of certain supplements can slow the biological processes of aging to a significant degree by slowing the rate of cell division and thus the rate of telomere shortening of somatic cells, by reducing free radical exposure and boosting antioxidant defenses, by modulating the activity of certain key enzymes (e.g. aromatase and 5-alpha Reductase enzymes), through supplementation with melatonin and oral growth hormone secretagogue supplements, and by maintaining a fasting blood sugar levels at or below 90 mg/dl.